FORMOSA CRYPTO

Completing the Chain:

Verified Implementations of Hash-Based Signatures and Their Security

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GitHub Repository

Paper

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XMSS: Overview

- XMSS [BDH11] is a stateful hash-based signature scheme.
- Security relies on the security of the underlying hash function.
- XMSS combines a one-time signature scheme (WOTS^+) with a Merkle tree.
  - The root of the Merkle tree is the public key of the XMSS instance.
  - Each leaf corresponds to a (compressed) WOTS^+ public key.
  - A Merkle tree of height h has 2^h leaves, and thus allows signing 2^h messages.

XMSS: Overview

- **Key generation:** Generate one-time signing keys and build a Merkle tree from their public keys. The tree root forms part of the XMSS public key.
- **Signing:** Sign the message digest with the next unused one-time key and include its authentication path in the signature. Update the signing index to prevent key reuse.
- **Verification:** Recover the one-time public key from the signature and use the authentication path to reconstruct the tree root. If it matches the root in the XMSS public key, then the signature is valid.
- **TreeHash:** An algorithm used to compute the root of a subtree by repeatedly hashing pairs of nodes, starting from its leaves.

Jasmin [ABB+17, ABB+20]

- High-level assembly language designed for cryptography implementations
  - High-level zero-cost abstractions: variables, arrays
  - Control over low-level details (e.g. instruction selection, loop unrolling, and whether variables are stored on the stack or in registers)
- Applicability to the development of high-speed, high-assurance cryptographic primitives has been demonstrated through applications to several primitives, most notably SHA-3 [ABB+19] and ML-KEM [AOB+24, ABB+23].
- Suported architectures: x86_64, ARMv7M, ARMv8-A, RISC-V

Jasmin [ABB+17, ABB+20]

- Stack zeroization [OBG+24]
- (Static) Safety checker
- Compiler correctness proved in Rocq
- Extraction to EasyCrypt
- Speculative constant-time:
  - Spectre v1 [SBG+23] and other spectre variants [ABC+25]
  - Compilation preserves SCT [ABB+25].

Compiler Correctness & Constant-Time Preservation

Compilation preserves the semantics of *safe* source programs and constant-time behavior: functional correctness and constant-time guarantees established at source level carry to the assembly level.

EasyCrypt

- Interactive proof assistant used for cryptographic security proofs and reasoning about program correctness.
- Supports several logics:
  - **Ambient Logic (AL):** structures proofs and combines results from the other logics.
  - **Hoare Logic (HL):** proves properties of a single program using preconditions and postconditions.
  - **Probabilistic Hoare Logic (pHL):** proves bounds on the probability of a program satisfying a postcondition.
  - **Probabilistic Relational Hoare Logic (pRHL):** proves equivalences between two probabilistic programs.

Verification Workflow

EasyCrypt

Jasmin

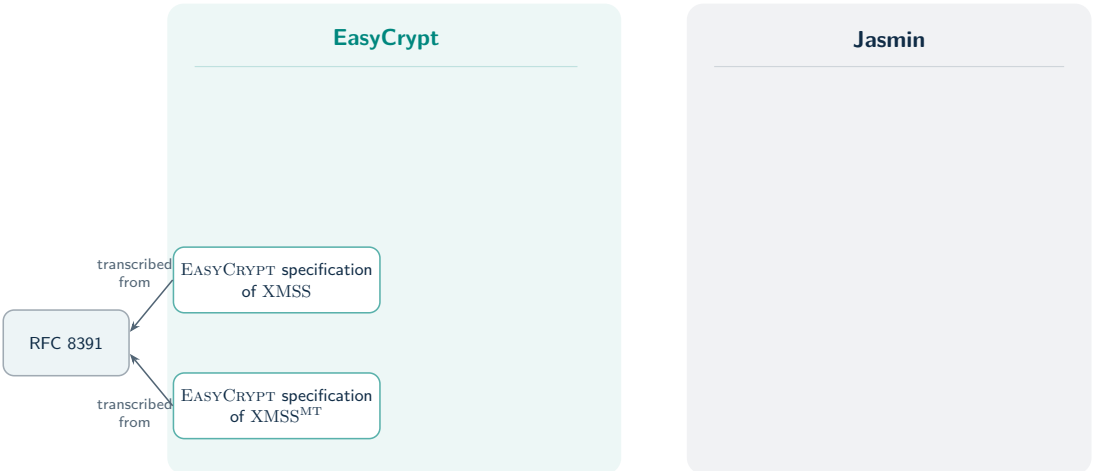
Verification Workflow

RFC 8391

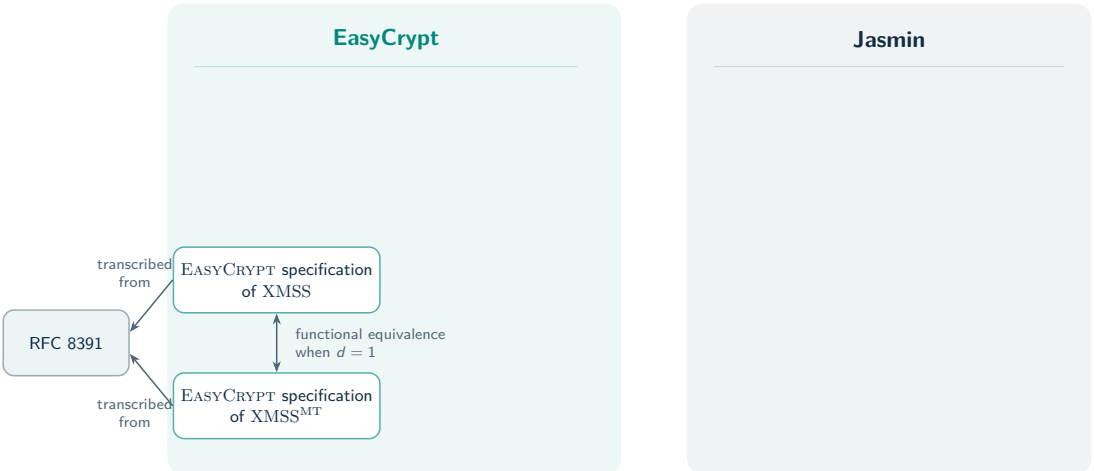
EasyCrypt

Jasmin

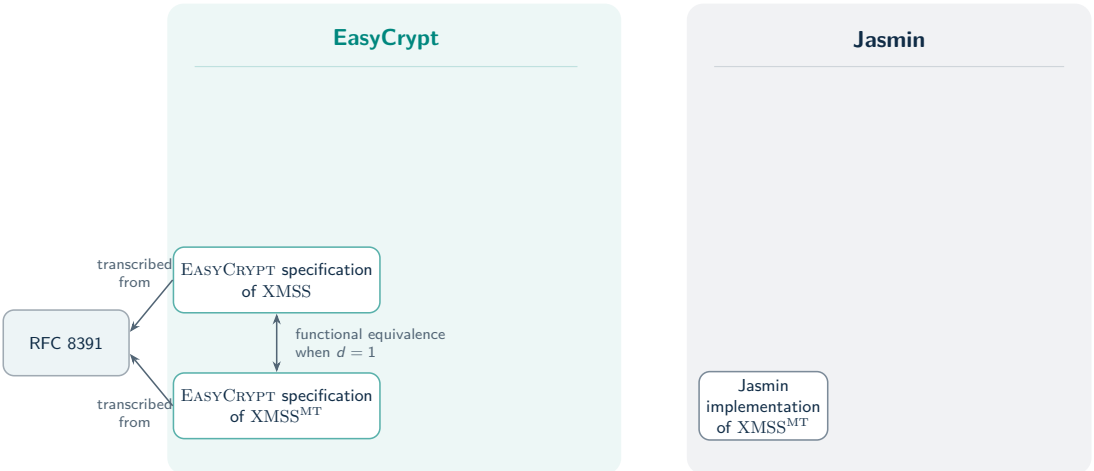
Verification Workflow



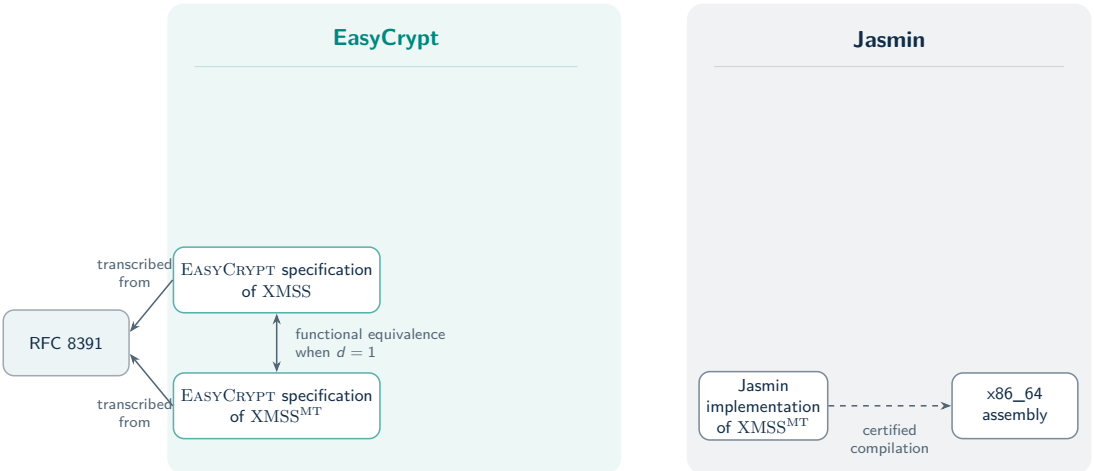
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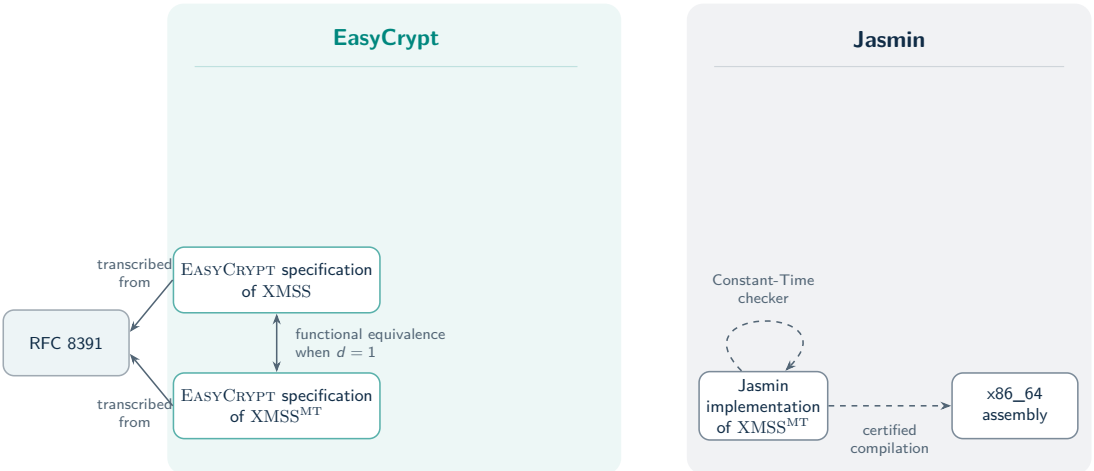
Verification Workflow



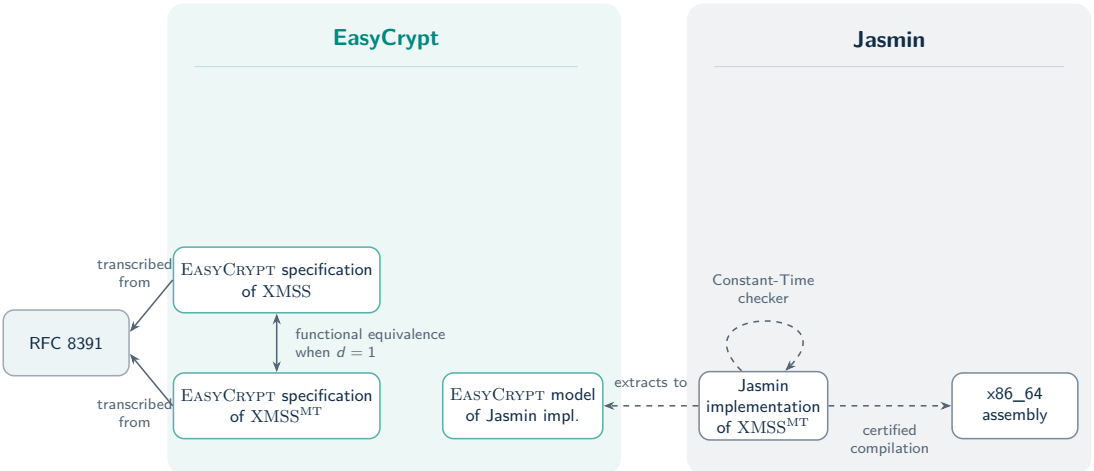
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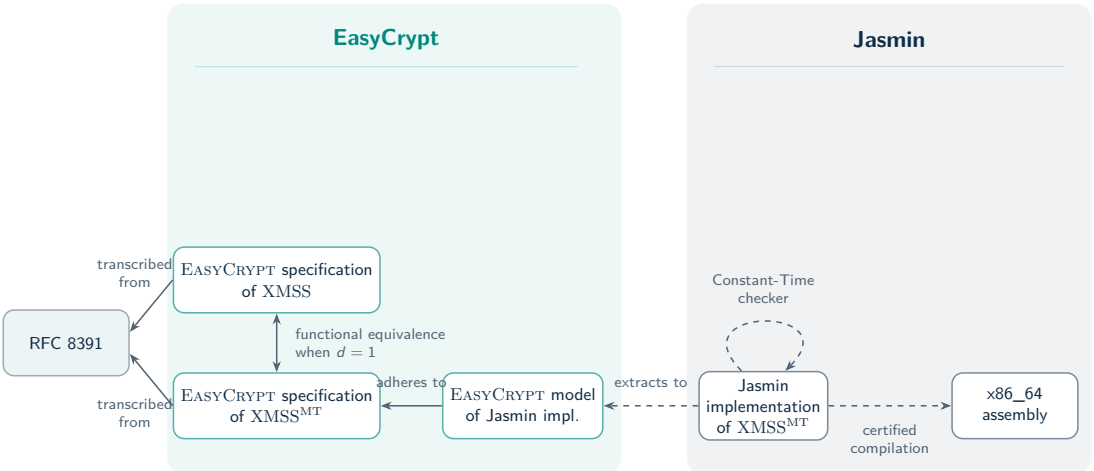
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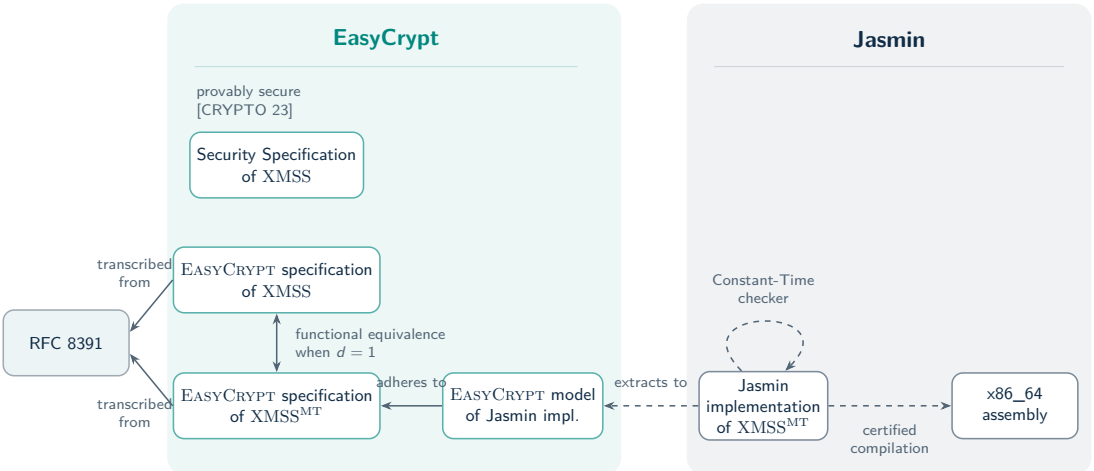
Verification Workflow



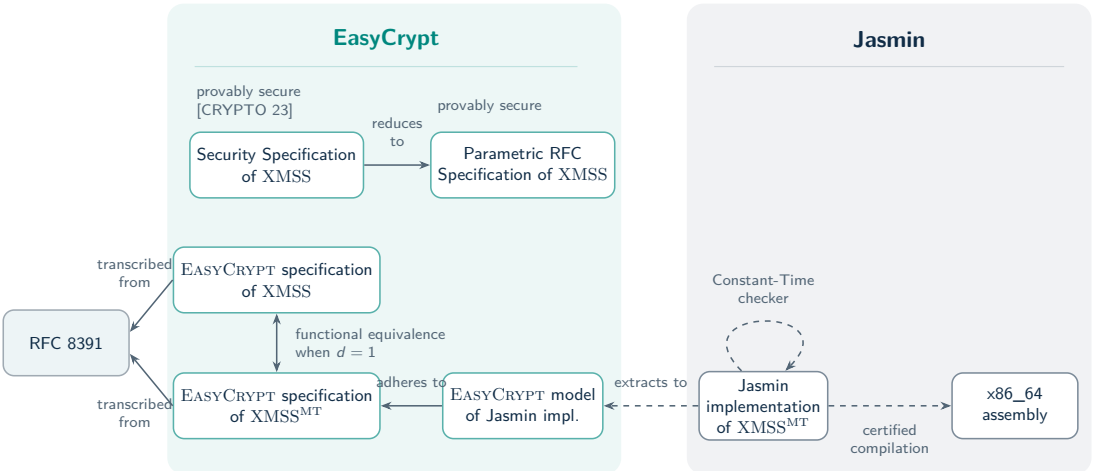
Verification Workflow



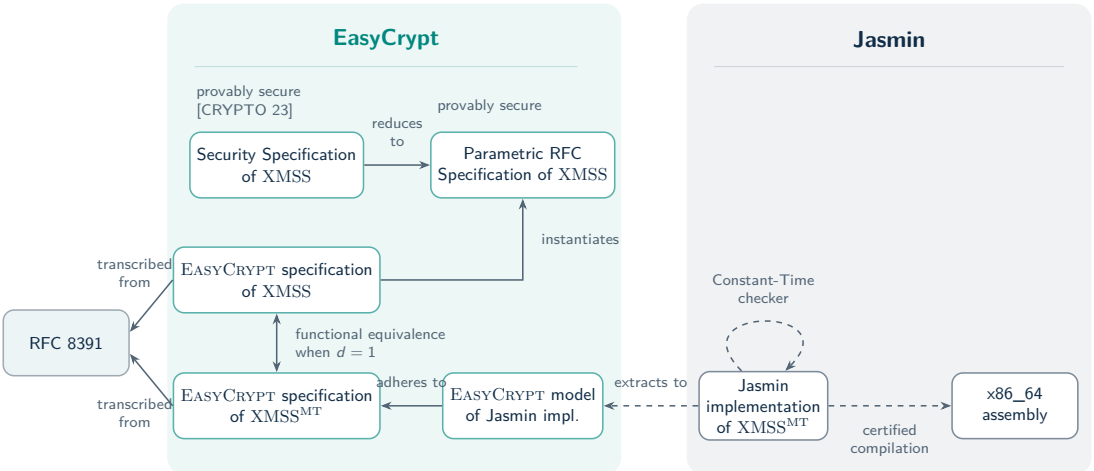
Verification Workflow



Verification Workflow



Verification Workflow



Trusted Computing Base

- Correctness of the SHA-256 implementation is outside the scope of the proof
- Jasmin compiler: unverified OCaml code (parsing and typing)
- Jasmin extraction to EasyCrypt
- EasyCrypt codebase
- SMT Solvers (Z3, CVC4 and Alt-Ergo)

Implementation: Overview

- $\mathrm{XMSS}^{\mathrm{MT}}$ implementation written in Jasmin targeting x86_64, based on the C reference implementation[1], prioritizing readability and closeness to RFC 8391.
- **SHA-256 from libjade**[2] (the C reference uses OpenSSL).
- **Fixed-size data:** stack arrays with sizes known at compile time; a separate hash function for each input length.
- **Variable-size data:** raw pointers with explicit loads and stores, e.g. for the message to be signed.
- **Arrays:** array arguments must be disjoint.

[1] https://github.com/XMSS/xmss-reference
[2] https://github.com/formosa-crypto/libjade

Implementation: Authentication Paths

- **TreeHash algorithm:** Used to compute the root of a subtree: keep intermediate nodes on a stack; merge the top two nodes whenever they have the same height.
- **Jasmin impl. follows RFC 8391:** compute each node of the authentication-path with a separate TreeHash call on the corresponding subtree.
- **C reference:** compute the root and authentication path together in a single TreeHash traversal.
- **Tradeoff**: Closer correspondence to the RFC simplifies the correctness proof, but increases execution time.

Implementation: Constant-Time and Zeroization

- **Constant-time validated with Jasmin's CT checker.** Annotate public values loaded from memory (e.g. message length): memory loads are secret by default, and need to be explicitly marked as public.
- **Stack zeroization on return from exported functions:** compiler-generated code clears the stack, ensuring that no secret data remains on the stack after function return.
  - Loop zeroization adds less than 0.4% overhead for all three operations. Unrolling adds about 30 KB of code and is slightly slower in these benchmarks.

Functional Correctness Proof

- **Goal**: Prove that the Jasmin implementation of $\mathrm{XMSS}^{\mathrm{MT}}$ adheres to the RFC-like EasyCrypt specification.
- Extract an EasyCrypt representation of the Jasmin program using the Jasmin compiler. Program equivalence is then expressed via **pRHL** statements.
- We assume the functional correctness of the underlying SHA-256 implementation.

Functional Correctness: Relating Representations

- **Specification:** records for keys and signatures.
- **Implementation:** byte arrays and raw pointers.
- Encoding and decoding operators connect these representations

```
lemma xmss_kg_correct :
    n = XMSS_N /\ (* remaining parameters ommited *) =>
    equiv [
      M(Syscall).__xmssmt_core_keypair ~ XMSS_MT_PRF.kg :
      true
      ==>
      res{1}.`1 = DecodePkNoOID res{2}.`2 /\
      res{1}.`2 = DecodeSkNoOID res{2}.`1
    ].
```

Figure: EasyCrypt lemma stating the correctness of the key generation algorithm.

Functional Correctness Proof: Reasoning About Memory

- Raw pointers require explicit reasoning about address arithmetic and the contents of memory (e.g. pointers don't overlap and do not overflow the 64-bit address space).

```
pred valid_ptr_i (p : W64.t) (o : int) =
  0 <= o =>
    0 <= to_uint p /\ to_uint (p) + o < W64.modulus.

pred disjoint_ptr (p1 l1 p2 l2 : int) =
  forall (k1 : int), 0 <= k1 < l1 =>
    forall (k2 : int), 0 <= k2 < l2 =>
      p1 + k1 <> p2 + k2.
```

Figure: Pointer predicates in EasyCrypt.

TreeHash

- Proving functional correctness of TreeHash was the most challenging part of the proof
- Computes the root of a subtree from its leaves:
  - **Security Specification**: Expressed as a recursive function over an inductive data type for trees.
  - **RFC Specification**: Iterative function that makes use of a stack for storing intermediate nodes.
- Rather than proving that implementation and specification are accessing the stack within bounds, we prove that they perform identical accesses (which could be within bounds or out of bounds), and prove that they produce the same result.

QUESTIONS?

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